

Bringing Space Weather Down to Earth

A Risk Communication Analysis

Elizabeth Pauley and Sarah Pawlowski | Advisor: Dr. Ann-Marie Reinhold



Introduction

Goal

Create a simplified space weather warning system that uses a single numerical risk scale, straight forward language, and conveys key insights through a dashboard.

What is space weather?

- Consists of particles and magnetic fields generated by the Sun
- Can impact Earth's upper atmosphere and magnetic field
- Affects Earth technology (e.g., satellites, power grids, radio communications, GPS, and aviation)
- Risks are comparable to other well-known major natural disasters, however:
 - Severe events are rare
 - Satellites are often difficult, if not impossible, to repair
 - People have limited options to prevent damage



Fig 1. Current warning and forecasting system (NOAA)

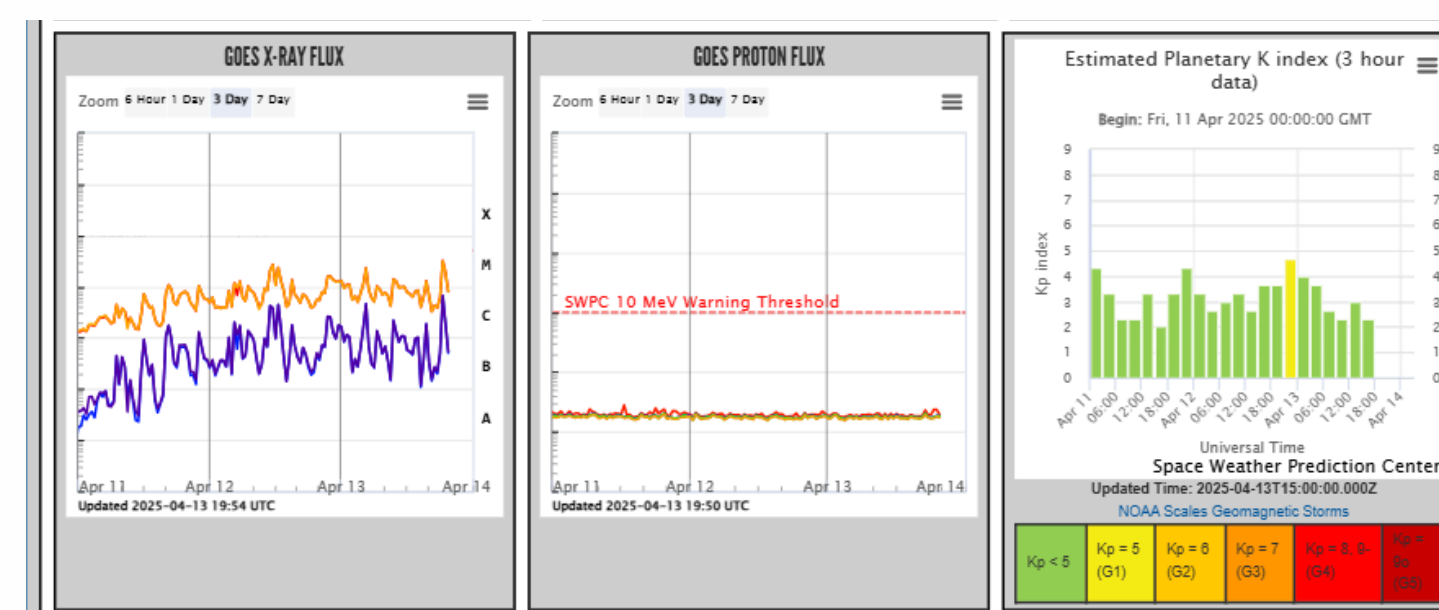


Fig 2. Current NOAA dashboard (NOAA)

- R** (Radio Blackouts) - when a solar X-ray flare changes the Earth's ionosphere, blocking high-frequency radio transmissions on the sunlit side of the Earth
- S** (Solar Radiation Storms) - an influx of high-energy particles from a solar event that can have negative impacts on space hardware
- G** (Geomagnetic Storms) - geomagnetic disturbances of the Earth's magnetic field that can primarily impact power grids and spacecraft operations

Language Analysis Results

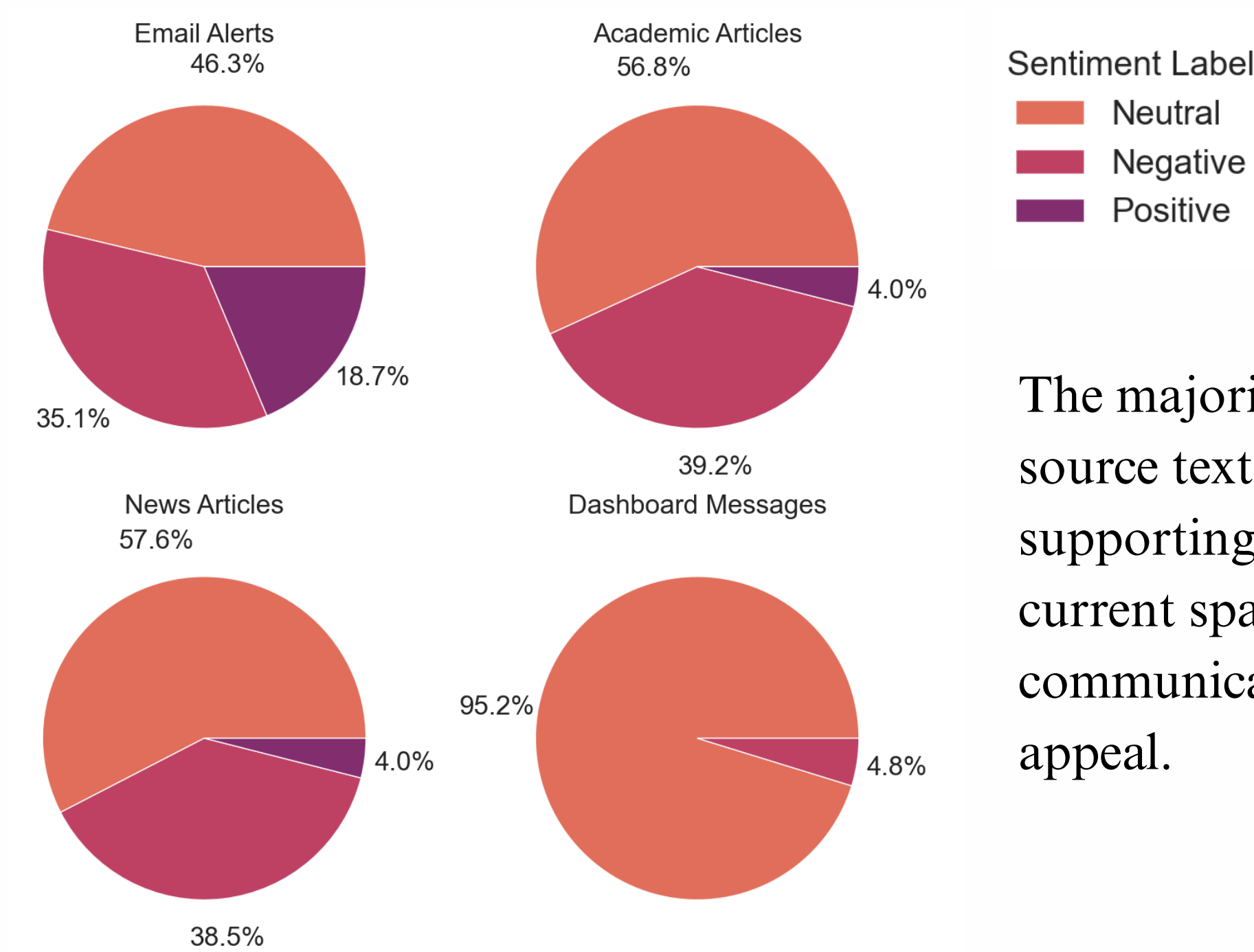


Fig 3. Sentiment proportions across text types

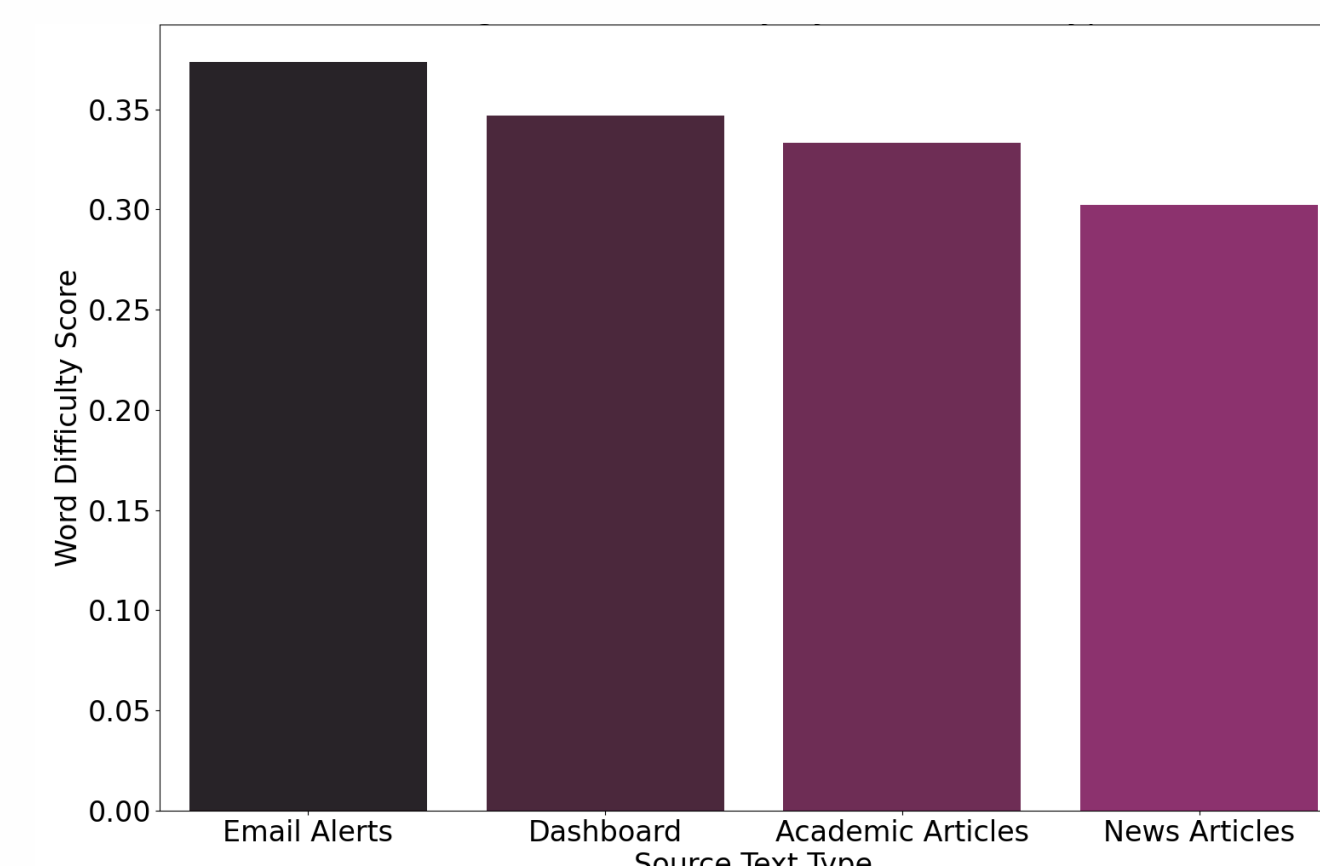


Fig 4. Average word difficulty across text types

The majority of sentiment for all source text types is neutral, supporting our hypothesis that current space weather communications lack emotional appeal.

Email alerts have the highest word difficulty average while news articles have the lowest.

The language used in our dashboard has a lower average word difficulty than email alerts which are currently the primary form of communication relating to space weather conditions.

Space Weather Dashboard

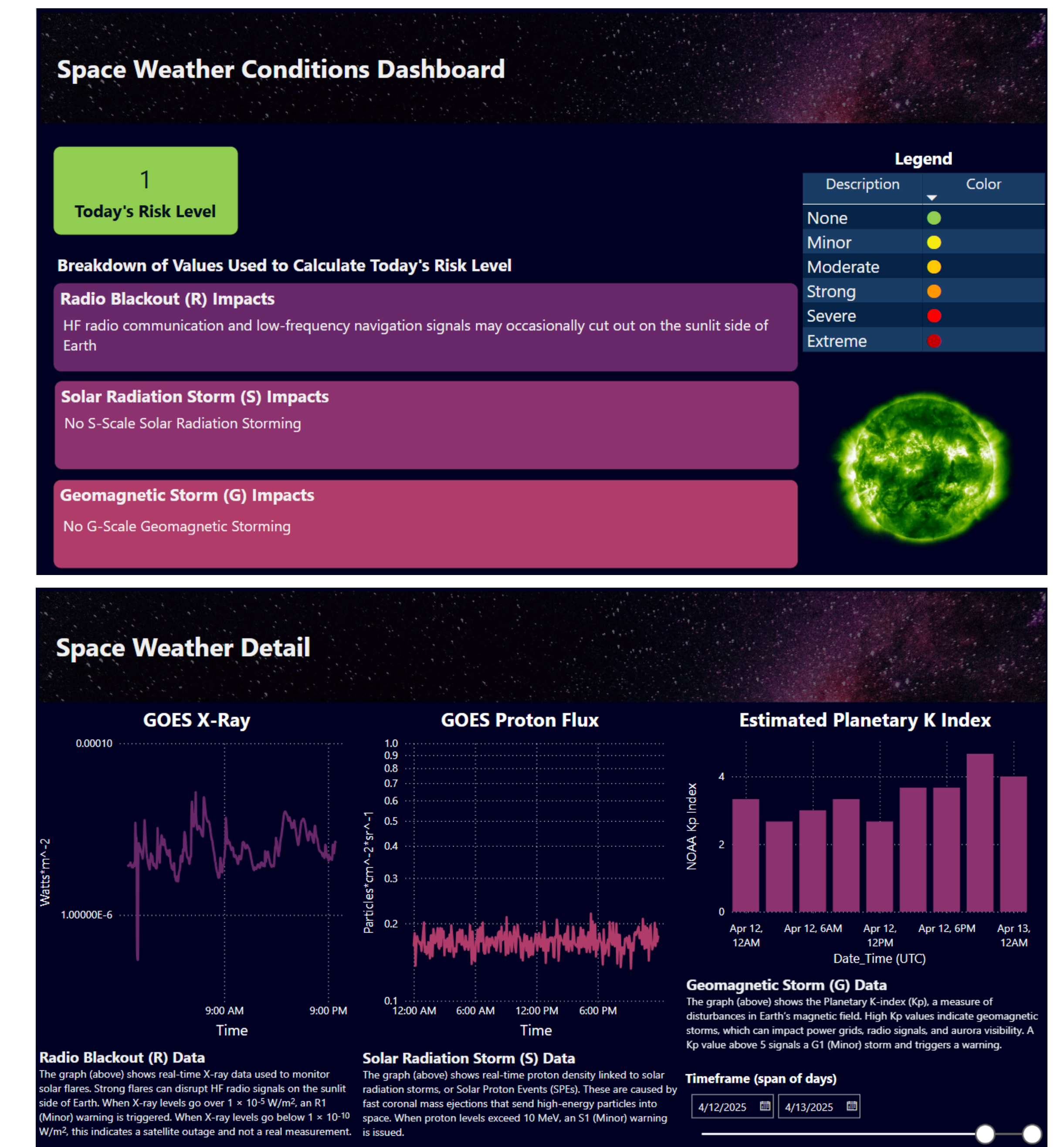


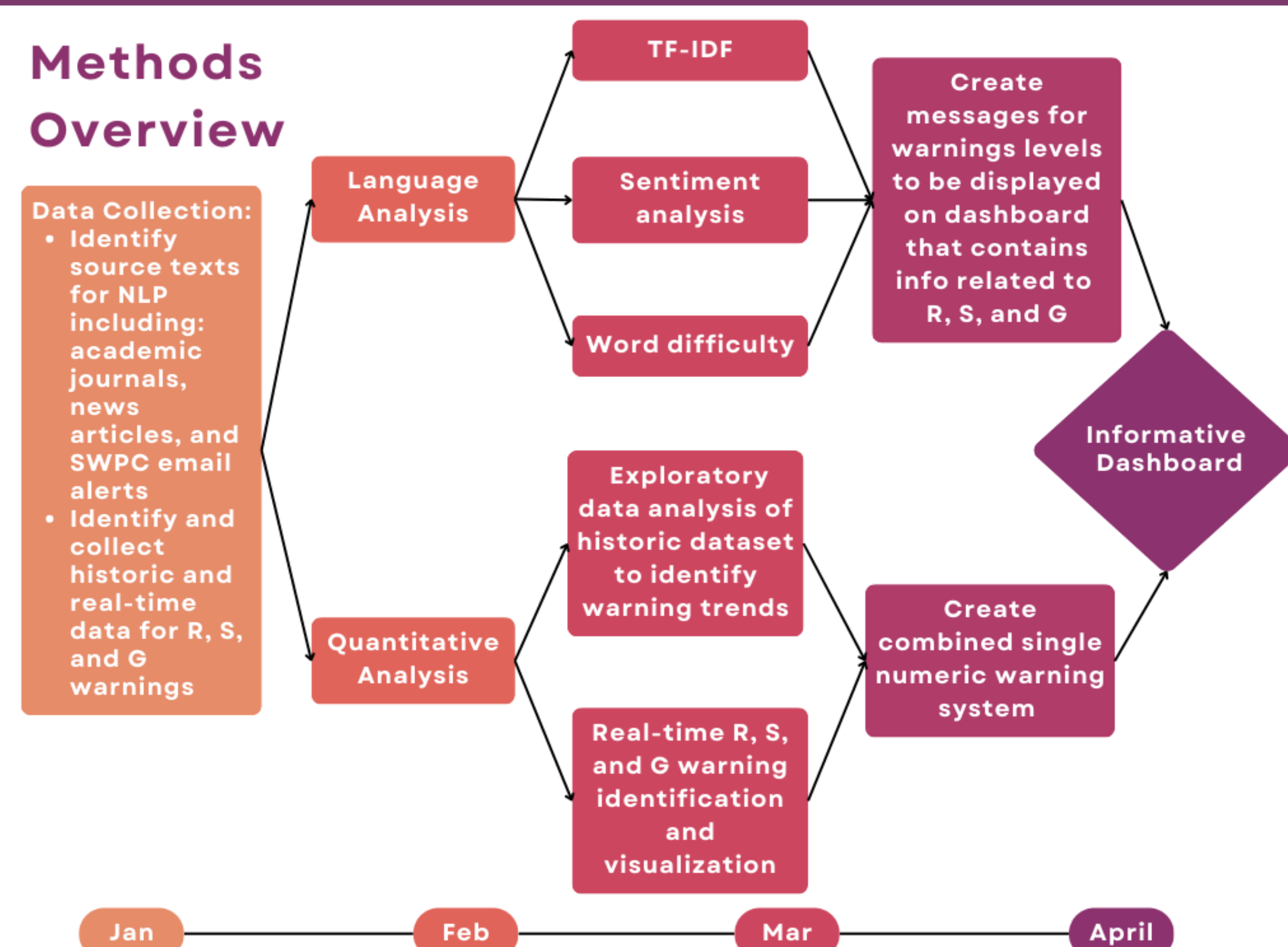
Fig 6. New and improved dashboard

Improvements included in our dashboard:

- The dashboard features a more simplistic and visually appealing interface
- R, S, and G values are combined to determine a single numerical risk value
- All graphs directly relate to R, S, and G warning levels, are displayed on a single page, and have clear explanations accompanying the data
- Impacts of R, S, and G events are explained in easy-to-understand terms
- Dashboard updates every 3 hours, providing more timely overall warning compared to NOAA

Methods

Methods Overview



Quantitative Analysis Results

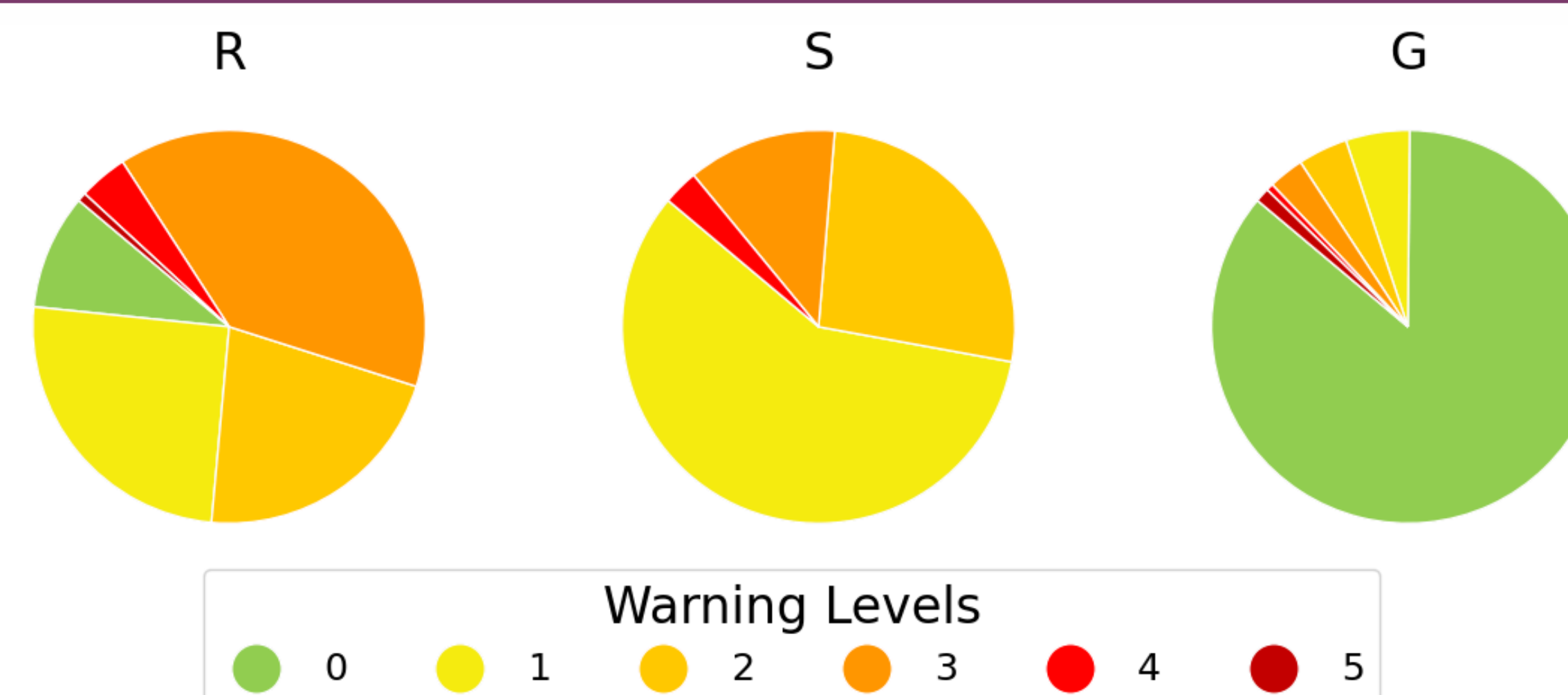


Fig 5. Historic R, S, and G warning ratios

There is minimal overlap between the three warning systems; they differ in severity, frequency, timing, and are not strongly correlated with each other. Therefore, to avoid eliminating critical information, our single numeric warning level is based on the highest of the R, S, and G values from the real-time data displayed on our dashboard. While this approach may overstate the risk to certain hazards, it captures the information required to remain informed of space weather threats.

References

- Homepage | NOAA / NWS Space Weather Prediction Center. NOAA.gov, 2000, www.swpc.noaa.gov/.
- Carr, C., & Steenburgh, R. (2020). NOAA Space Weather Prediction Center, Solar Proton Events Affecting the Earth Environment, 1976 - Present. National Centers for Environmental Information. <https://www.ngdc.noaa.gov/stp/space-weather/interplanetary-data/solar-proton-events/SEP%20page%20code.html>
- Duskin, Omer. "Estimating Word Difficulty in English Using Machine Learning (ML)." Medium, Python in Plain English, 20 Nov. 2023, python.plainenglish.io/estimating-word-difficulty-in-english-using-machine-learning-ml-5d366c0f0700.